HANZHE ZHANG'S RESEARCH STATEMENT

Updated: 09/10/26; Latest: http://hanzhezhang.github.io/research/HanzheResearchStatement.pdf

I am an applied microeconomic theorist who develops and uses price-theoretic and game-theoretic tools to study important economic and social phenomena. I test my theories empirically and experimentally and apply them to a wide range of interdisciplinary, classroom, and public settings.

In the last few years, I have published in leading economics journals such as *Journal of Political Economy*, *Review of Economic Studies*, *Journal of Economic Theory*, *Games and Economic Behavior*, *American Economic Journal: Microeconomics*, and *Journal of Development Economics*, and leading interdisciplinary journals such as *Proceedings of the National Academy of Sciences*, *Journal of Financial and Quantitative Analysis*, and *Sustainable Cities and Society*; and my research has been financially supported by leading public and private entities such as NSF, Amazon, and Anthropic.

My interests include (a) matching and bargaining, especially with applications to labor economics and family economics, (b) evolutionary and psychological game theory, and (c) mechanism design (such as auctions) in commercial and financial settings. These interests continue my PhD thesis "Essays in matching, auctions, and evolution" under Gary Becker and Phil Reny at the University of Chicago. In addition, I received financial support and garnered media attention for my research in (d) science of science and innovation and (e) interdisciplinary settings. This document focuses on my publications, working papers, and ongoing projects in the last few years.

A. Matching and bargaining

My primary topic of interest is two-sided matching, whose successful applications include the National Resident Matching Program for medical students, school choice, and kidney exchanges. My recent studies include (1) assortative mating, (2) the bargaining aspects of matching markets, and (3) the diverse effects of the marriage market on individuals and the society.

A.1. Assortative matching (*PNAS* & *JPE*)

Characterizing the extent of assortative mating—partnership and marriage of like types—have interested researchers across many fields—economics, sociology, psychology, biology, to a name a few. I study how likely individuals sort on biological and phenotypical characteristics such as height and weight as well as on socioeconomic characteristics such as education and income.

In [16] ("Assortative mating on blood type: Evidence from China," *Proceedings of the National Academy of Sciences*, December 2022), my coauthors and I use 1 million Chinese couples to find strong and robust evidence for assortative mating on blood type—one of the most fundamental phenotypes in biological, medical, and psychological studies. To establish the results, we use a set of existing measures of assortative mating in the literature.

However, theoretically, it is unclear what would be the most appropriate measure, and different measures may lead to different conclusions. For example, Eika, Mogstad, and Zafar (2019) and Chiappori, Costa Dias, and Meghir (2020) use different measures and find opposing conclusions about whether assortative mating on education has increased in the US over the recent decades. I take an *axiomatic* approach: start with a set of properties (i.e., axioms) that the measures should satisfy and establish how appropriate the measures are for capturing changes in assortative matching through the lenses of these properties. I find that under natural axioms (e.g., scale invariance, monotonicity, type invariance, and decomposability), the normalized trace (the percentage of pairs who share the same type) is the unique measure that conveys the ordinal and cardinal contents of assortative mating. I also provide characterization results for the commonly used measures of the odds ratio and aggregate likelihood ratios.

My primarily theoretical investigation is merged with the primarily empirical investigation of Chiappori, Costa Dias, and Meghir (2020) and is published as the lead article in the *Journal of Political Economy*, one of the top five journals in economics ([26] "Changes in martial sorting: Theory and evidence from the US," October 2025). The axiomatic approach to study assortative matching and other socioeconomic outcomes such as intergenerational mobility has gained popularity and is becoming the norm rather than the exception in the literature, evident in recent working papers such as McGee (2026) and Mohan and Vilalta-Bufi (2026).

A.2 Bargaining and bargaining in matching markets (*REStud*; *AEJMicro*; *AEJMicro*)

Before they match, individuals negotiate on the amount they transfer to each other (e.g., dowries in marriage markets, salaries in labor markets, and contracts in labor and financial markets). I study these bargaining processes and outcomes.

To start, I study bilateral bargaining problems. My coauthor and I study reputational bargaining when outside options are present ([19] "Reputational bargaining with ultimatum opportunities," *Review of Economic Studies*, July 2025). Two parties negotiate in the presence of external resolution opportunities (e.g., court, arbitration, or war). The outcome of external resolution depends on the privately held justifiability/strength of their claims. A justified party issues an ultimatum for resolution whenever possible, but an unjustified party strategically bluffs with an ultimatum to establish a reputation for being justified. We show that the availability of external resolution opportunities can benefit or hurt an unjustified party in equilibrium. When the chances of being justified become negligible, agreement is immediate and efficient; and if the set of justifiable demands is rich, our solution modifies the Nash-Rubinstein bargaining solution of Abreu and Gul (2000) in a simple way.

Shapley and Shubik (1972), Becker (1973), and Kelso and Crawford (1982) pioneered the so-called transferable utilities matching model that provides predictions on matching and bargaining outcomes. It has been the workhorse model of several of my earlier work ([6] "Pre-matching gambles," *Games and Economic Behavior*, May 2020; [9] "An investment-and-marriage model with differential fecundity: On the college gender gap," *Journal of Political Economy*, May 2021; and [17] "A marriage-market perspective on risk-taking and career choices," *European Economic Review*, February 2023).

Whether individuals match and negotiate the transfers in real life as modeled had not been thoroughly tested. To study how the model performs, I collaborate with experimentalists to examine the theoretical predictions of the matching model ([18] "Decentralized matching with transfers: Experimental and noncooperative analyses," *American Economic Journal: Microeconomics*, November 2024). We provide one of the first comprehensive experimental and theoretical investigations of decentralized matching markets with transfers. Our contributions are threefold. First, our experiment shows that matching patterns and aggregate surpluses are affected by whether efficient matching is assortative and whether equal splits are sustainable; in the canonical theory, neither is predicted to matter. Second, in markets with equal numbers of participants on both sides, individual payoffs in our and others' experiments cannot be explained by existing refinements of the core but are consistent with our noncooperative model's unique prediction. Third, in markets with unequal numbers of participants on the two sides, subjects on the long side of the market do not compete fully; these noncompetitive outcomes are not captured by the canonical theory, but by our noncooperative model.

In addition, the canonical model is static and involves one-to-one matching. In real world, many markets are dynamic and involve many-to-one matching. For example, in labor markets, firms are long-lived, workers are short-lived, and firms may hire several workers at once, who may produce synergy among them and may also have preferences for whom they work with. It has been known that complementarities and peer effects in static many-to-one markets lead to nonexistence of stable matchings and consequently potentially market chaos. My coauthors and I show that in dynamic markets, when firms are sufficiently patient, stability is restored and guaranteed, even with complementarities and peer effects ([28] "Self-enforced job matching," *American Economic Journal: Microeconomics*). Wage flexibility (i.e., transferable utility) is crucial to our result, as it not only facilitates surplus extraction when firms cooperate in no-poaching agreements but also enhances the threat of punishment through bidding wars.

A.3. Marriage and migration (*JDE*)

Continuing from my earlier work, I examine the role of the marriage market in affecting various individual decisions and consequently the society. In earlier work, I examine how the marriage market affects risk-taking [6]; college investment and marriage age [9]; and career investments [17]. In recent work, I examine the role of the marriage market in migration decisions.

In [22] ("Young women in cities: Urbanization and gender-biased migration," *Journal of Development Economics*, January 2025), my empirical coauthors and I attribute documented gender differences in migration to marriage incentives. Observationally, young women outnumber young men in cities in many countries during periods of economic growth and urbanization. This gender imbalance among young urbanites is more pronounced in larger cities. We use the gradual rollout of Special Economic Zones across China as a quasi-experiment to establish the causal impact of urbanization on gender-differentiated incentives to migrate. We highlight the role of the marriage market in increasing rural women's chance of marrying and marrying up in urban areas during rapid urbanization. This paper was covered by VoxChina.

B. Evolutionary and psychological game theory (*JET*; *AEJMicro r&r*; *GEB*)

I study the intergenerational and long-run effects of the marriage market on preference evolution, which contrasts with the short-run effects considered in the papers described above. This line of research combines my interests in matching and evolutionary economics. In [11] ("Preference evolution in different marriage markets," *European Economic Review*, June 2021), we examine preference evolution under different marriage market arrangements when preferences are influenced by parental and own choices. When everyone has homophilic preferences (that is, they prefer to marry partners with the same traits), assortative matching is critical in determining cultural evolution. When some individuals are heterophilic (preferring individuals with different cultural traits), the analysis is technically challenging, especially when combined with different channels of intergenerational transmission. In [17] ("Heterophily, stable matching, and intergenerational transmission in cultural evolution," *Journal of Economic Theory*, June 2023), we characterize different cultural processes in settings with different combinations of (i) homophilic and heterophilic marital preferences, (ii) a men-optimal or women-optimal stable matching scheme, and (iii) familial and societal forces of intergenerational transmission. We show that, in the presence of even a small fraction of heterophilic individuals, there is cultural homogeneity in which only one cultural trait survives. Cultural heterogeneity (diversity) arises only when proposers are all homophilic or all members of a cultural group are homophilic.

In our recent paper [30] "A justification for equal sharing" (*American Economic Journal: Microeconomics*, revise and resubmit), we take the evolution one step further. We study the evolution of productive traits under different surplus-sharing norms in one-sided matching markets with heterogenous agents. Agents form partnerships endogenously, and the output of each partnership depends on the productivity of the matching agents. The division of output is governed by a social norm that specifies how partners share surplus. Our results provide an evolutionary rationale for equal sharing norms.

I explain, from evolutionary and psychological perspectives, behaviors that deviate from rationality. I have examined a wide range of topics: overconfidence [7], non-Bayesian updating [3], and the promotion of green building practices [15]. Recently, in [22] "Pay it forward: Theory and experiment," *Games and Economic Behavior*, October 2025, we investigate the psychological motivations behind pay-it-forward behavior: individuals are more likely to forward kindness to a third party when they have been treated kindly. We find that altruism and indirect reciprocity spur people to pay kind actions forward, which informs how kindness begets further kindness; inequity aversion limits the amounts given, even when an individual knows that her kindness will be paid forward. Our paper elucidates how kind behaviors get passed on between parties that never directly interact, which has implications for the formation of social norms and behavior conduct within workplaces, neighborhoods, and communities.

C. Mechanism design in financial and commercial settings (*JFQA* and working papers)

Another topic of my interest is when and how auctions and related selling mechanisms should be used. In [23] ("Borrowing stigma and lender of last resort policies", *Journal of Financial and Quantitative Analysis*, February 2025), my finance coauthor and I study when the lender of last resort—usually the central bank—should use auctions to provide liquidity to banks during periods of financial crisis. During the 2008 financial crisis, banks avoided borrowing from the Federal Reserve's long-established discount window (DW). In contrast, they actively participated in its special temporary monetary program, the Term Auction Facility (TAF), even though the programs had the same borrowing requirements. We explain how the two programs suffer from different stigma and how the introduction of TAF incentivized banks' borrowing. We synthesize several data sources to confirm our theory, whereby banks that borrow from the TAF are financially stronger than those that borrow from the DW. The paper was recognized as the best paper in annual meetings of the Midwest Finance Association and of Eastern Finance Association.

In [33] "When visibility shapes timing: Evidence on bidding dynamics and market efficiency," a major online labor-market platform abruptly switched from open auctions to sealed auctions. My coauthors in information systems and I find that, after the switch, workers bid 42.5% faster and ask for 16.5% lower wages. The auction duration decreases by 38.9%, and the probability of a successful hire increases by 6.1%. Though employers choose from a 10.3% reduced pool of participating bidders, there is no evidence of a significantly different wage for the chosen workers. Post-project outcomes also improve. These findings highlight the advantages of sealed auctions in improving market efficiency, match quality, and participation retention in online labor markets. In contrast, in open auctions, workers have strong incentives to delay bidding or place initial bids higher than their valuation to learn from competitors' bids and adjust accordingly.

In [34] "Join or wait? Seller capability development and dynamic platform pricing," my coauthors and I rationalize why e-commerce platforms support and subsidize suppliers in rural areas to manufacture and sell products online. We argue that, despite platforms' philanthropic claims, these actions are strategically designed to enhance profitability. By providing early subsidies to young sellers, platforms incentivize entry and reduce learning costs, later recouping these investments as sellers gain experience and increases sales. Using a dynamic model of two-sided markets, we analyze intertemporal and cross-side pricing strategies of a platform with market power. Our findings indicate that sellers' network externalities and learning-by-doing effects reinforce each other, motivating the platform to subsidize them. This study bridges two typically distinct areas of the economy: global online platforms and less developed rural regions.

D. Science of science and innovation (working papers)

A new strand of my research in the last few years, supported by an Amazon grant "Predicting successful scientific collaboration" since 2023, is the emerging field of science of science of innovation, the study of researchers and scientists.

In [29] "High and rising institutional concentration of award-winning economists," we document and attempt to explain a high and rising institutional concentration of award winners in economics, compared to 17 other fields in natural sciences, engineering, and social sciences. We collect nearly 300,000 annual affiliations of nearly 6,000 award-winning researchers. All fields, except for economics, exhibit a low and decreasing concentration. We investigate potential reasons for the anomaly, including researcher mobility, reliance on physical assets, the age of fields, the role of prestige, and geographic influence. The findings have been featured in major outlets, e.g., The Atlantic, Financial Times, Forbes, and Guardian. I was one of the keynote speakers at the International Conference on the Science of Science and Innovation in the National Academy of Sciences and my coauthor presented it at the NBER.

In [31] "Over a century of economics research collaboration," we study collaboration patterns among 100,000+ authors using bibliographic information of nearly 500,000 economics publications and working papers since 1886. We document several new patterns. First, inter-institutional collaboration was U-shaped: declining through the 20th century and rising in the 21st. Second, the experience composition of research teams—the mix of junior and senior economists—has remained stable. Third, returns to collaboration have shifted in waves: solo-author papers were most likely to be highly cited in the 1950s, two-author papers in the 1960s-1980s, three-author papers in the 1990s-2000s, and four-plus-author papers in the 2010s. Using COVID as a natural experiment that shifts collaboration costs, we find a polarizing effect: some researchers retreated to solo work while others formed larger teams. We develop a parsimonious theoretical framework of skill complementarity, coordination costs, and heterogeneous responses to cost shocks to rationalize all empirical findings.

Artificial intelligence is the hottest topic now. My coauthors and I study how AI and human brain interact and whether innovation will get easier. In [32] "Artificial intelligence and the brain: Is innovation getting easier?" we treat AI as a synthesizer of existing knowledge into refined knowledge. The human brain then recombines it to generate new ideas. By lowering the cognitive burden of information processing, AI can accelerate discovery, but it may also reduce knowledge spillovers by filtering out valuable information. We show that research productivity varies nonmonotonically with AI efficiency: AI boosts innovation when knowledge is very scarce or very abundant. Yet, it may create a mid-knowledge-level *AI trap* in which faster AI progress slows down innovation and lowers productivity.

E. Interdisciplinary projects (*JBG*; *CRC*; *SCS*)

I concluded a $1.44 million NSF project as a co-PI in 2024 and co-produced several papers in construction management: [24] "How homebuyers' cognition and affect shape their attitude on green buildings," *Journal of Green Building*, June 2025; [20] "Iterative development of dynamic student project team interventions", *Construction Research Congress*, March 2024; and [18] "Profit versus sustainability in bikeshare," *Sustainable Cities and Society*, June 2023.

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Chiappori, Pierre-André, Monica Costa Dias, and Costas Meghir, 2020. "Changes in assortative matching: Theory and evidence for the US." Working Paper no. 26932, NBER, Cambridge, MA.
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My Papers

All publications are peer-reviewed. In most papers, authors follow economics convention to be equally contributing first authors ordered alphabetically. When noted by **, authors follow conventions in interdisciplinary fields to be ordered according to contribution.

<u>Working papers</u>

[34] Join or wait? Seller capability development and dynamic platform pricing,
Kevin Mi, Danxia Xie, Buyuan Yang, and Hanzhe Zhang,
September 2026.

[33] When visibility shapes timing: Evidence on bidding dynamics and market efficiency,
Qiang Gao, Chenhui Guo, Mingfeng Lin, and Hanzhe Zhang,
August 2026.

[32] Artificial intelligence and the brain: Is innovation getting easier?
Kevin Mi, Danxia Xie, Buyuan Yang, and Hanzhe Zhang,
June 2026.

[31] Over a century of economics research collaboration,
Willy Chen, Xiao Qiao, and Hanzhe Zhang,
June 2026.

[30] A justification for equal sharing,
Jiabin Wu and Hanzhe Zhang,
American Economic Journal: Microeconomics, revise and resubmit, March 2026.

[29] High and rising institutional concentration of award-winning economists,
Richard B. Freeman, Danxia Xie, Hanzhang Zhou, and Hanzhe Zhang,
July 2024.
Media coverage: The Atlantic, Financial Times, Firstpost, Forbes, Forum New Economy, Guardian.

<u>Since the previous reporting period from January 2023</u>

[28] Self-enforced job matching,
Ce Liu, Ziwei Wang, and Hanzhe Zhang,
American Economic Journal: Microeconomics, May 2026.

[27] Pay it forward: Theory and experiment,
Amanda Chuan and Hanzhe Zhang,
Games and Economic Behavior, October 2025.
[26] Changes in marital sorting: Theory and evidence from the US,
Pierre-André Chiappori, Monica Costa Dias, Costas Meghir, and Hanzhe Zhang
Journal of Political Economy, lead article, October 2025.
[25] Reputational bargaining with external resolution opportunities
Mehmet Ekmekci and Hanzhe Zhang,
Review of Economic Studies, July 2025.
[24] How homebuyers' cognition and affect shape their attitude on green buildings,
**Yongsheng Jiang, Zihao Xu, Jiayu Chen, Dong Zhao, Hanzhe Zhang, and Ying Li,
Journal of Green Building, accepted, June 2025.
[23] Borrowing stigma and lender of last resort policies,
Yunzhi Hu and Hanzhe Zhang,
Journal of Financial and Quantitative Analysis, 60(1): 374–405, February 2025.
[22] Young women in cities: Urbanization and gender-biased migration,
Yumi Koh, Jing Li, Yifan Wu, Junjian Yi, and Hanzhe Zhang,
Journal of Development Economics, special issue on "Migration and Development," 172, 103378, January 2025.
[21] Decentralized matching with transfers: Experimental and noncooperative analyses,
Simin He, Jiabin Wu, Hanzhe Zhang, and Xun Zhu,
American Economic Journal: Microeconomics, 16(4): 406–439, November 2024.
[20] Iterative development of dynamic student project team interventions,
**Arnav Jain, Faizan Shafique, Sinem Mollaoglu, Xu Dong, Hanzhe Zhang, Shimeng Dai, Kenneth Frank, Dorothy Carter, Anna Young Argyris, Annick Anctil, and Kristen Cetin,
Construction Research Congress, March 2024.

<u>Not considered in the previous reporting period</u>

[19] Marital preferences and stable matching in cultural evolution (with Victor Hiller and Jiabin Wu),
Journal of Economic Theory, 120, 105671, June 2023.
[18] Profit versus sustainability in bikeshare,
Huiyi Litan, Ke Rong, Youran Wu, Danxia Xie, Hanzhe Zhang, and Dong Zhao,
Sustainable Cities and Society, special issue on "Sustainable Cities and Modern Multi-Energy Societies," 93, 104512, June 2023.
[17] A marriage-market perspective on risk-taking and career choices,
Hanzhe Zhang and Ben Zou,
European Economic Review, 152, 104379, February 2023.

<u>Considered in the previous reporting period for tenure evaluation (August 2015 to December 2022)</u>

[16] Assortative mating on blood type: Evidence from one million Chinese pregnancies
Yao Hou, Ke Tang, Jingyuan Wang, Danxia Xie, and Hanzhe Zhang
Proceedings of the National Academy of Sciences, December 2022.
[15] Decision model to optimize long-term subsidy strategy for green building promotion
**Yongsheng Jiang, Tao Hu, Dong Zhao, Bingsheng Liu, Hanzhe Zhang, Yunjia Zhang, and Zihao Xu
Sustainable Cities and Society, 86, 104126, November 2022.
[14] Polarization, antipathy, and political activism
Jiabin Wu and Hanzhe Zhang,
Economic Inquiry, lead article, 600(3), 1005-1017, July 2022.
[13] Project team collaborations during time of disruptions: Transaction costs, knowledge flows, and social network theory perspective,
**Hasan Gokberk Bayhan, Sinem Mollaoglu, Hanzhe Zhang, and Kenneth Frank,
Construction Research Congress, 1012-1023, March 2022.

[12] Prices versus auctions in large markets,
Hanzhe Zhang,
Economic Theory, 72(4), 1297–1337, November 2021.
[11] Preference evolution in different matching markets,
Jiabin Wu and Hanzhe Zhang,
European Economic Review, 137, 103804, August 2021.
[10] Tractable cubic cost functions for teaching microeconomics,
Scott Swinton and Hanzhe Zhang,
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[9] An investment-and-marriage model with differential fecundity: On the college gender gap,
Hanzhe Zhang,
Journal of Political Economy, 129(5), 1464–1486, May 2021.
[8] The optimal sequence of prices and auctions,
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European Economic Review, 133, 103681, April 2021.
[7] An evolutionary justification for overconfidence,
Kim Gannon and Hanzhe Zhang,
Economics Bulletin, 40(3), 2494–2504, September 2020.
MSU University Undergraduate Arts & Sciences Forum (UURAF) best paper
[6] Pre-matching gambles,
Hanzhe Zhang
Games and Economic Behavior, 121, 76–89, May 2020.
Prior to reporting period
[5] The optimal sequence of prices and auctions (extended abstract)
Hanzhe Zhang,
Proceedings of the 3rd Conference on Auctions, Market Mechanisms and Their Applications, 36, August 2015.
[4] To risk or not to risk? Improving financial risk taking of older adults by online social information
**Jason Chen Zhao, Wai-Tat Fu, Hanzhe Zhang, Shengdong Zhao, and Henry Been-Lirn Duh
Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work and Social Computing, 95–104, March 2015.
[3] Evolutionary justifications for non-Bayesian beliefs
Hanzhe Zhang,
Economics Letters, 121, 198–201, November 2013.
[2] Two-player zero-sum poker models with one and two rounds of betting,
Hanzhe Zhang,
Penn Science, 9(1), 27-30, Fall 2010.
[1] Cohort profile: The China Jintan child cohort study,
**Jianghong Liu, Linda McCauley, Yang Zhao, Hanzhe Zhang, Jennifer Pinto-Martin, and Jintan Cohort Study Group.
International Journal of Epidemiology, 39(3), 668-674, June 2010.